pyWATS Queue Architecture Guide

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Version: 0.2.0+

Overview

The pyWATS queue system provides **reliable, priority-based queuing** for test reports with support for file persistence, crash recovery, and concurrent async processing. The architecture uses a **layered design** where base memory queues are extended for different use cases.

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Queue Hierarchy

Class Inheritance Structure

«abstract»
BaseQueue
+size property
+stats property
+add(data, item_id, priority)
+get(item_id)
+update(item)
+remove(item_id)
+list_pending()
+list_processing()
+list_completed()
+list_failed()
implements
MemoryQueue
-_items: Dict<str, QueueItem>
-_heap: List<QueueItem>
-_lock: threading.Lock
-_max_size: int
-_hooks: QueueHooks
+thread_safe operations
+add(data, item_id, priority)
+get(item_id)
+update(item)
+remove(item_id)
+get_next() : QueueItem
+list_by_status(status)
+clear_completed()
extends
PersistentQueue
Separate implementation
NOT extending MemoryQueue
File-based, async-first
AsyncPendingQueue
-_semaphore: asyncio.Semaphore
-_max_concurrent: int
-_observer: Observer
-_active_uploads: set
+concurrent_uploads
+watchdog_monitoring
+run() : async
+stop() : async
-_upload_report(file) : async
-_find_next_queued_file() : async

```
-_queue_dir: Path
-_delete_completed: bool
+file_persistence
+crash_recovery
---
+add(data, item_id, priority)
+update(item)
+remove(item_id)
-_save_to_disk(item)
-_load_from_disk()
-_delete_from_disk(item_id)
-_recover_crashed_items()
```

Design Principles

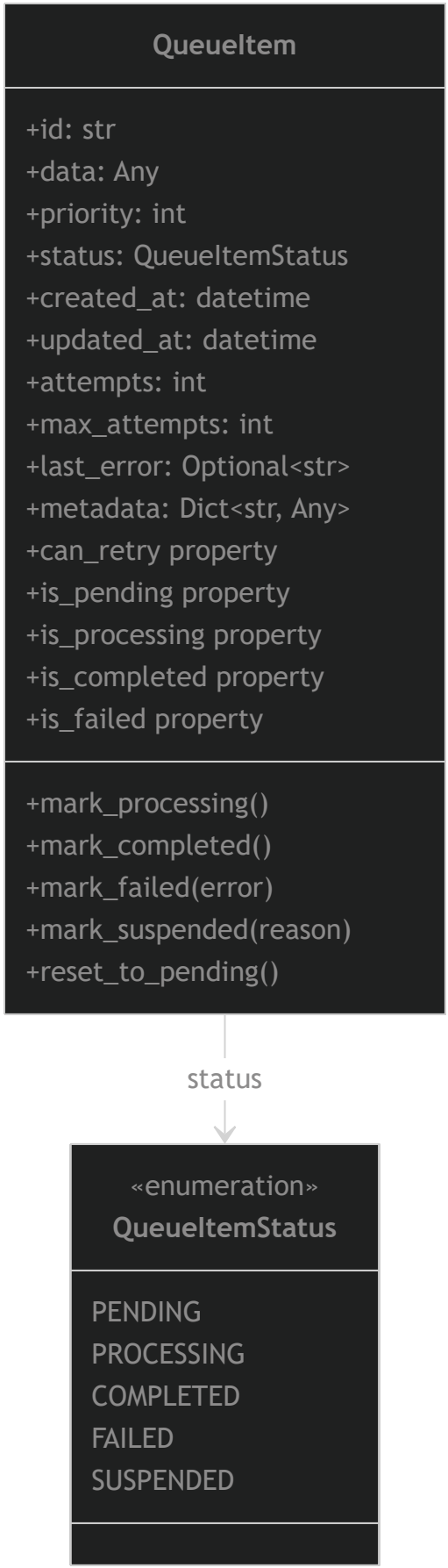
Separation of Concerns:

Layer	Package	Responsibility	File I/O
Abstract	`pywats.queue`	Define interface	❌ No
In-Memory	`pywats.queue`	Thread-safe queue ops	❌ No
Persistent	`pywats_client.queue`	File-based storage	✅ Yes
Async	`pywats_client.service`	Concurrent async uploads	✅ Yes

Key Principle: The core `pywats` API has **NO file operations** - it's pure in-memory. All file I/O lives in `pywats_client`.

QueueItem Data Model

Structure

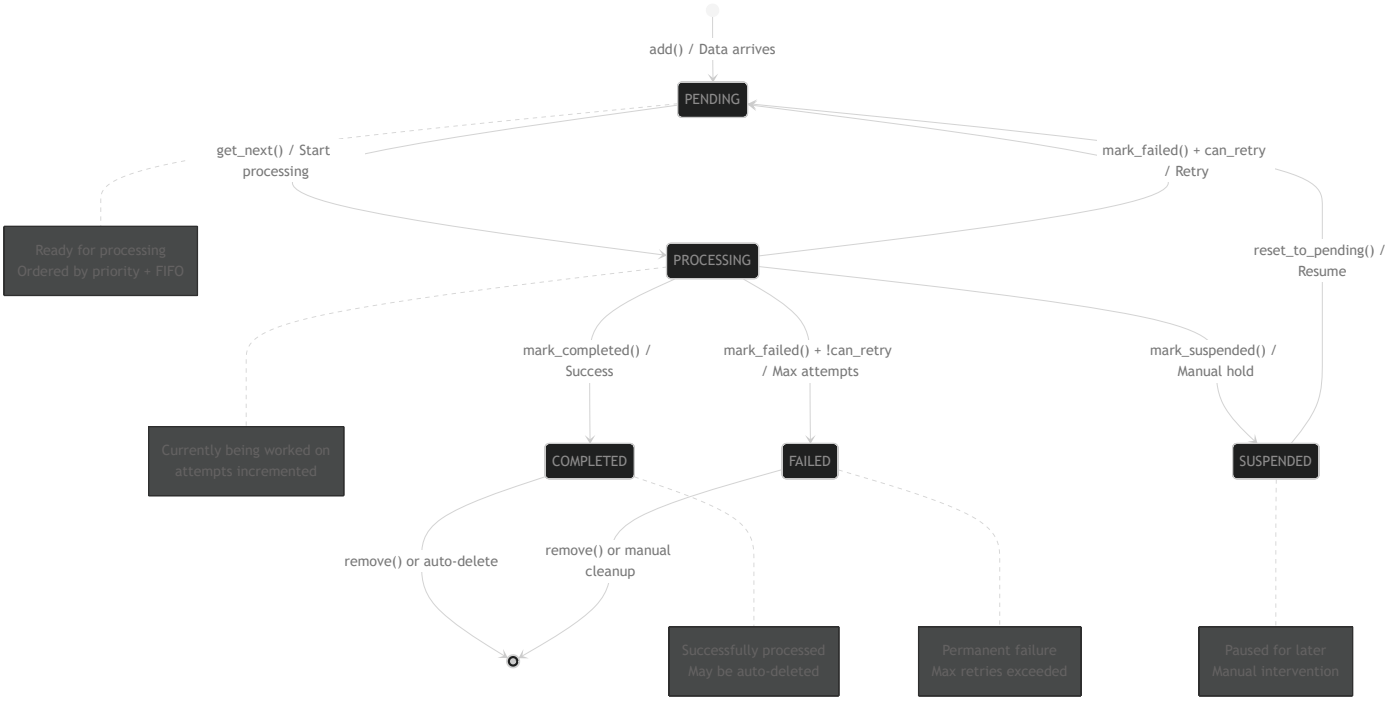

QueueItem
+id: str
+data: Any
+priority: int
+status: QueueItemStatus
+created_at: datetime
+updated_at: datetime
+attempts: int
+max_attempts: int
+last_error: Optional<str>
+metadata: Dict<str, Any>
+can_retry property
+is_pending property
+is_processing property
+is_completed property
+is_failed property
+mark_processing()
+mark_completed()
+mark_failed(error)
+mark_suspended(reason)
+reset_to_pending()
status
«enumeration»
QueueItemStatus
PENDING
PROCESSING
COMPLETED
FAILED
SUSPENDED

Field Responsibilities

Field	Purpose	Managed By
id	Unique identifier (UUID or custom)	User or auto-generated
data	Actual payload (report dict/model)	User
priority	Queue ordering (1=highest, 10=lowest)	User (default: 5)
status	Current processing state	Queue operations
created_at	Timestamp of creation	Auto-set
updated_at	Last status change	Auto-updated
attempts	Number of processing attempts	Auto-incremented
max_attempts	Retry limit	User (default: 3)
last_error	Last error message	Queue on failure
metadata	Custom key-value pairs	User

Queue Lifecycle States

State Diagram



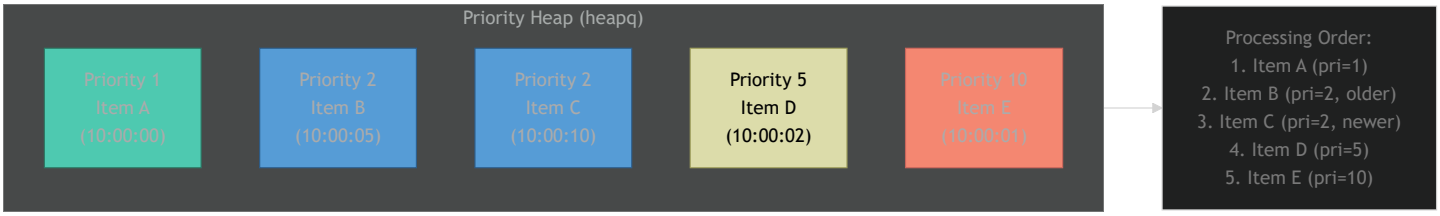
State Transitions

From	To	Trigger	Condition
`PENDING`	`PROCESSING`	`get_next()` → `mark_processing()`	Item is fetched for work
`PROCESSING`	`COMPLETED`	`mark_completed()`	Successfully processed
`PROCESSING`	`PENDING`	`mark_failed()`	`attempts < max_attempts`
`PROCESSING`	`FAILED`	`mark_failed()`	`attempts >= max_attempts`
`PROCESSING`	`SUSPENDED`	`mark_suspended()`	Manual hold
`SUSPENDED`	`PENDING`	`reset_to_pending()`	Resume processing
`COMPLETED`	*(deleted)*	`remove()` or auto-delete	Cleanup
`FAILED`	*(deleted)*	`remove()`	Manual cleanup

Priority Queue Mechanism

Priority Ordering

Rule: Items are processed in **priority order** (1=highest, 10=lowest), with **FIFO** within the same priority.



Comparison Logic:

```
def __lt__(self, other: "QueueItem") -> bool:
    # Compare (priority, created_at) tuples
    # Lower priority number = higher priority
    # Earlier timestamp = processed first (FIFO)
    return (self.priority, self.created_at) < (other.priority, other.created_at)
```

Use Cases

Priority	Use Case	Example
1	Critical/Emergency	Production line failure reports
2	High-priority	First article inspection
3	Above-normal	Engineering debug reports
5	Normal (default)	Standard production tests
7	Low-priority	Bulk historical uploads
10	Background	Archive/cleanup operations

MemoryQueue (Base Implementation)

Purpose

Pure in-memory queue with thread-safe operations. No file I/O - completely in-memory.

Responsibilities

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Key Operations

Operation	Complexity	Thread-Safe	Description
add()	O(log n)	✅ Yes	Add item to heap
get_next()	O(log n)	✅ Yes	Pop highest priority item
update()	O(n)	✅ Yes	Update item (no reheap)
remove()	O(n)	✅ Yes	Delete item by ID
list_pending()	O(n)	✅ Yes	Filter by status
size	O(1)	✅ Yes	Count all items

Design Note: `update()` does NOT re-heapify. If priority changes, item must be removed and re-added.

PersistentQueue (File-Based Storage)

Purpose

Extends MemoryQueue with file persistence and crash recovery.

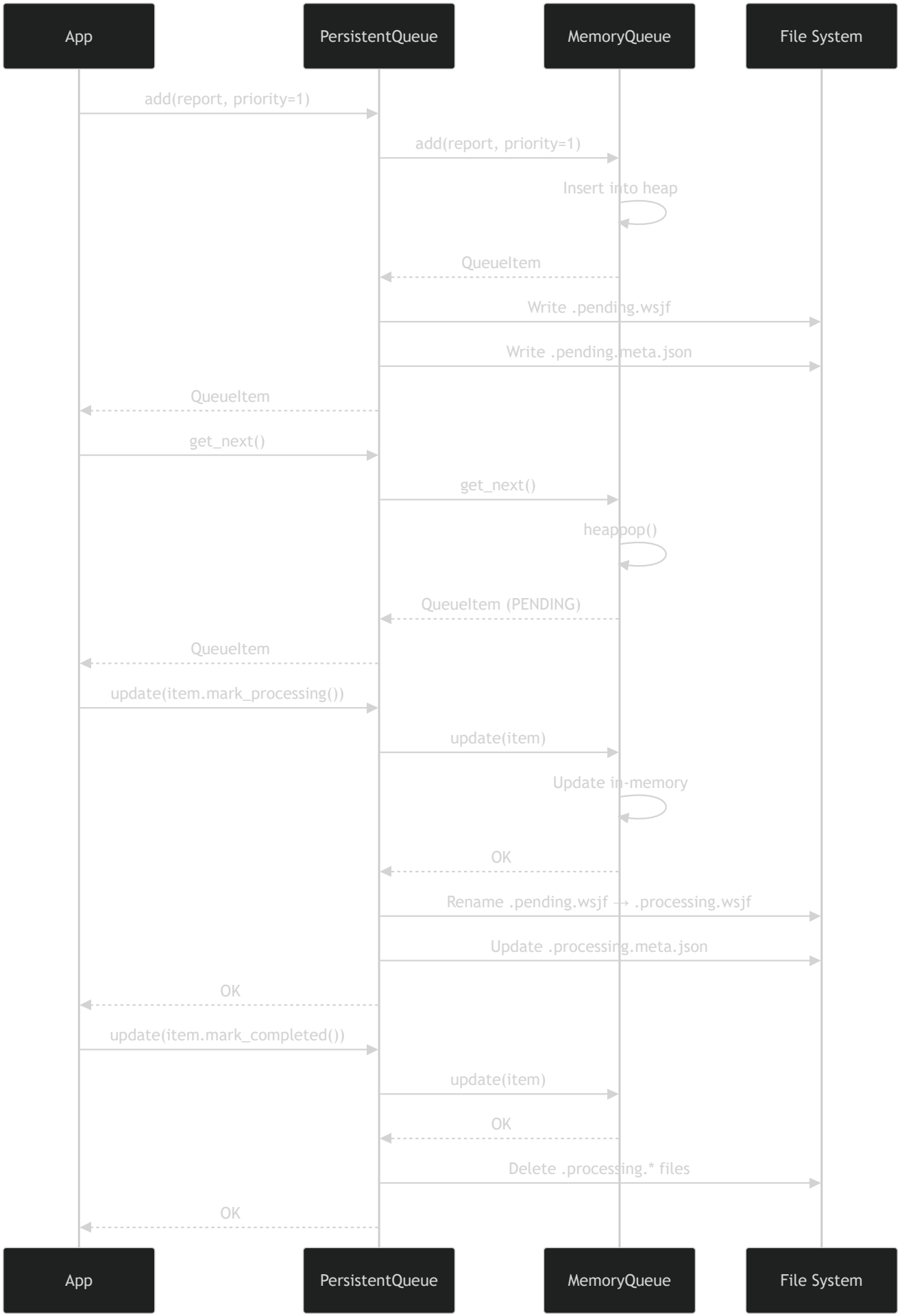
File Structure

```
C:\ProgramData\pyWATS\instances\{instance_id}\queue\
├── {report_id}.pending.wsjf          # Report data (PENDING)
├── {report_id}.pending.meta.json     # Metadata (attempts, errors)
├── {report_id}.processing.wsjf       # Report data (PROCESSING)
├── {report_id}.processing.meta.json  # Metadata
├── {report_id}.failed.wsjf           # Report data (FAILED)
├── {report_id}.failed.meta.json      # Metadata
└── {report_id}.completed.wsjf        # Report data (COMPLETED) - optional
```

File Extensions by Status:

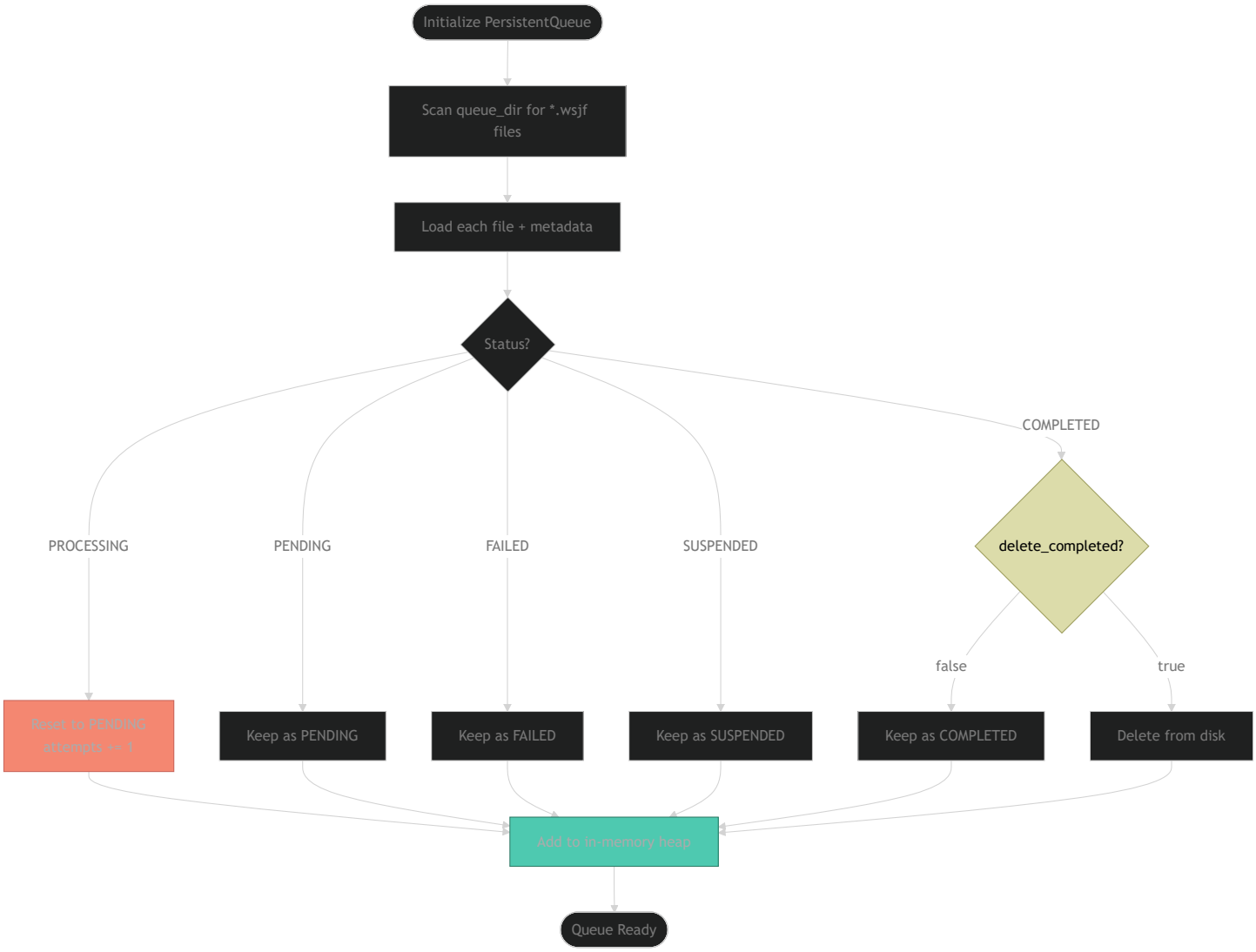
Status	Data File	Metadata File
PENDING	.pending.wsjf	.pending.meta.json
PROCESSING	.processing.wsjf	.processing.meta.json
COMPLETED	.completed.wsjf	.completed.meta.json
FAILED	.failed.wsjf	.failed.meta.json
SUSPENDED	.suspended.wsjf	.suspended.meta.json

Architecture


App
PersistentQueue
MemoryQueue
File System
add(report, priority=1)
add(report, priority=1)
Insert into heap
QueueItem
Write .pending.wsjf
Write .pending.meta.json
QueueItem
get_next()
get_next()
heappop()
QueueItem (PENDING)
QueueItem
update(item.mark_processing())
update(item)
Update in-memory
OK
Rename .pending.wsjf → .processing.wsjf
Update .processing.meta.json
OK
update(item.mark_completed())
update(item)
OK
Delete .processing.* files
OK
App
PersistentQueue
MemoryQueue
File System

Crash Recovery

On startup, PersistentQueue:



Recovery Logic:

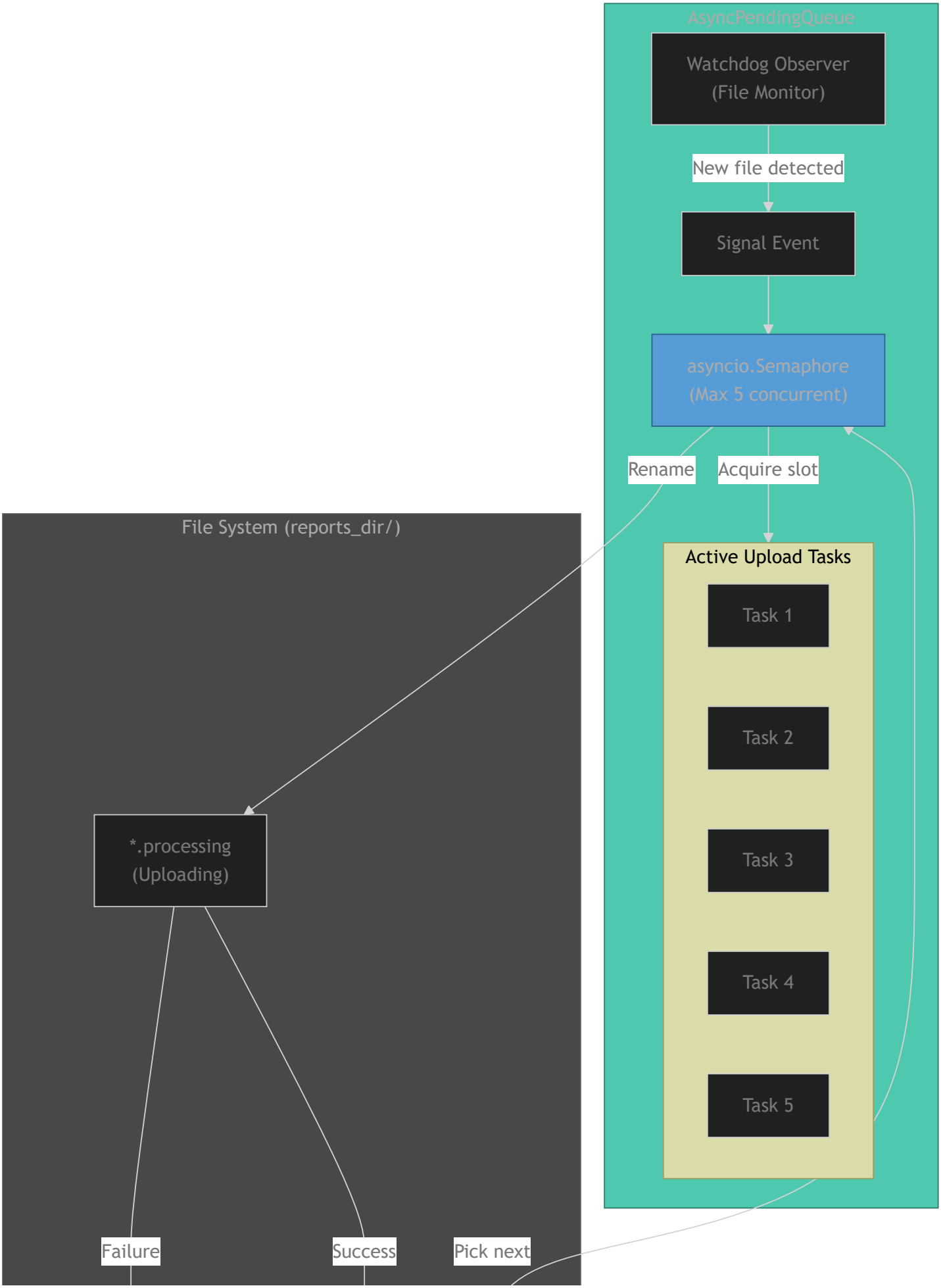
1. **PROCESSING** → Reset to **PENDING** (crash during upload)
2. **PENDING** → Keep as **PENDING** (ready to process)
3. **FAILED** → Keep as **FAILED** (permanent failure)
4. **COMPLETED** → Delete if `delete_completed=True`
5. **SUSPENDED** → Keep as **SUSPENDED** (manual hold)

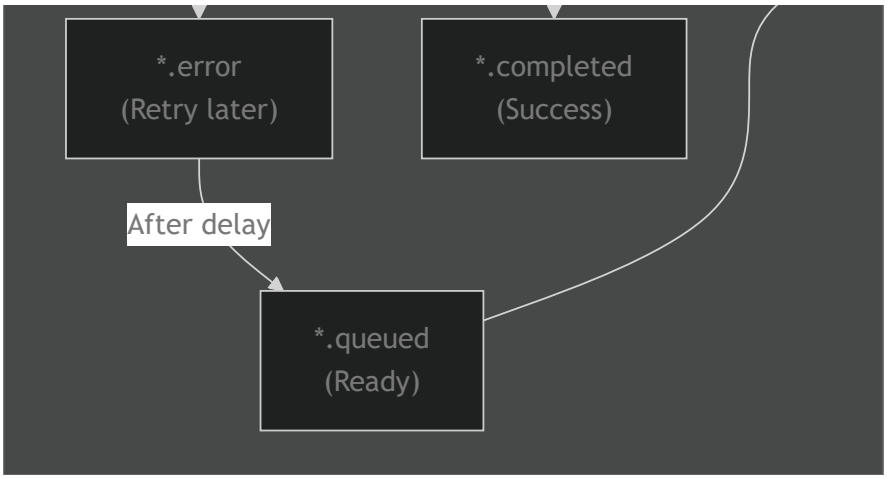
AsyncPendingQueue (Concurrent Async)

Purpose

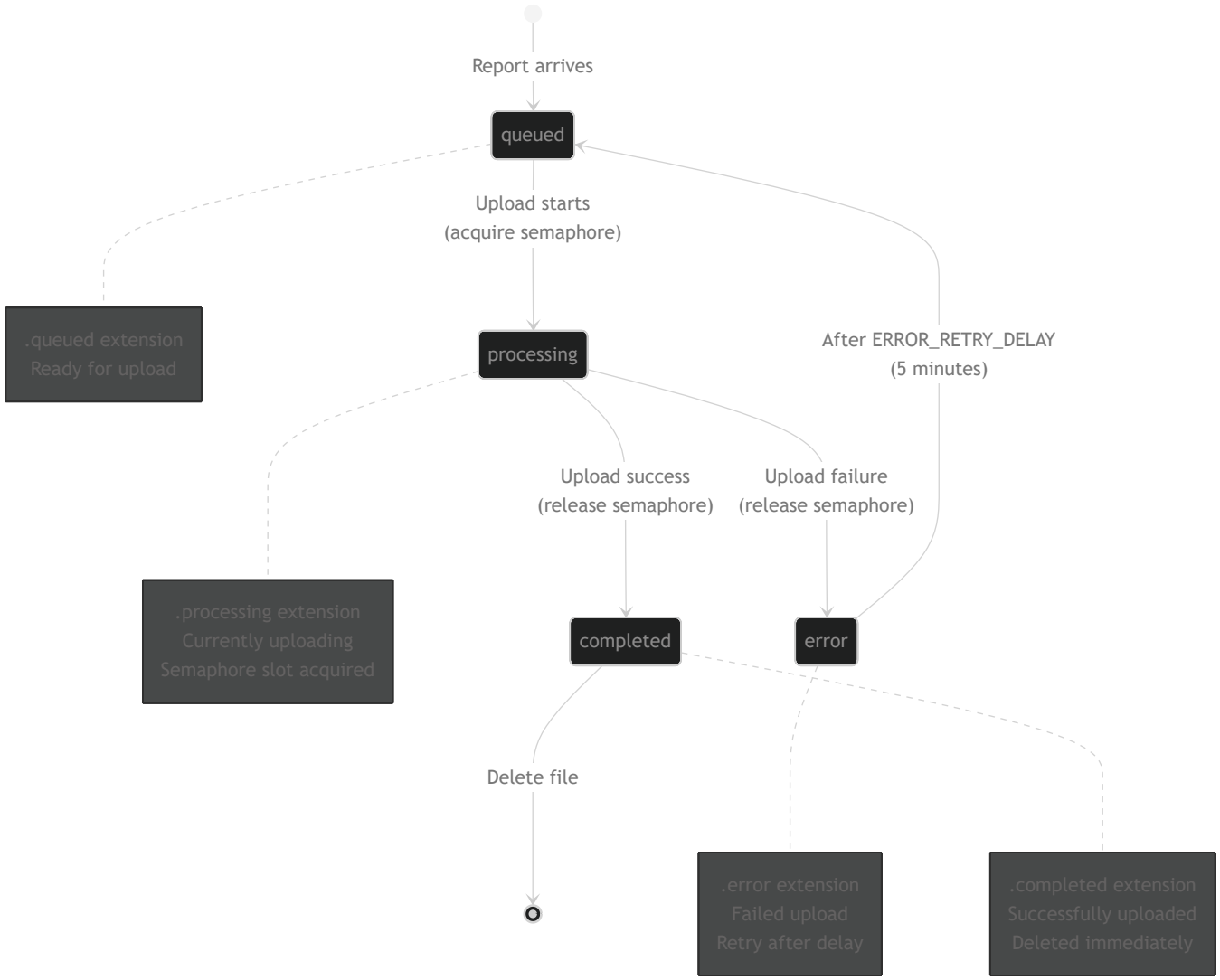
File-based async queue for concurrent report uploads with watchdog monitoring. **NOT** extending MemoryQueue - separate implementation.

Architecture

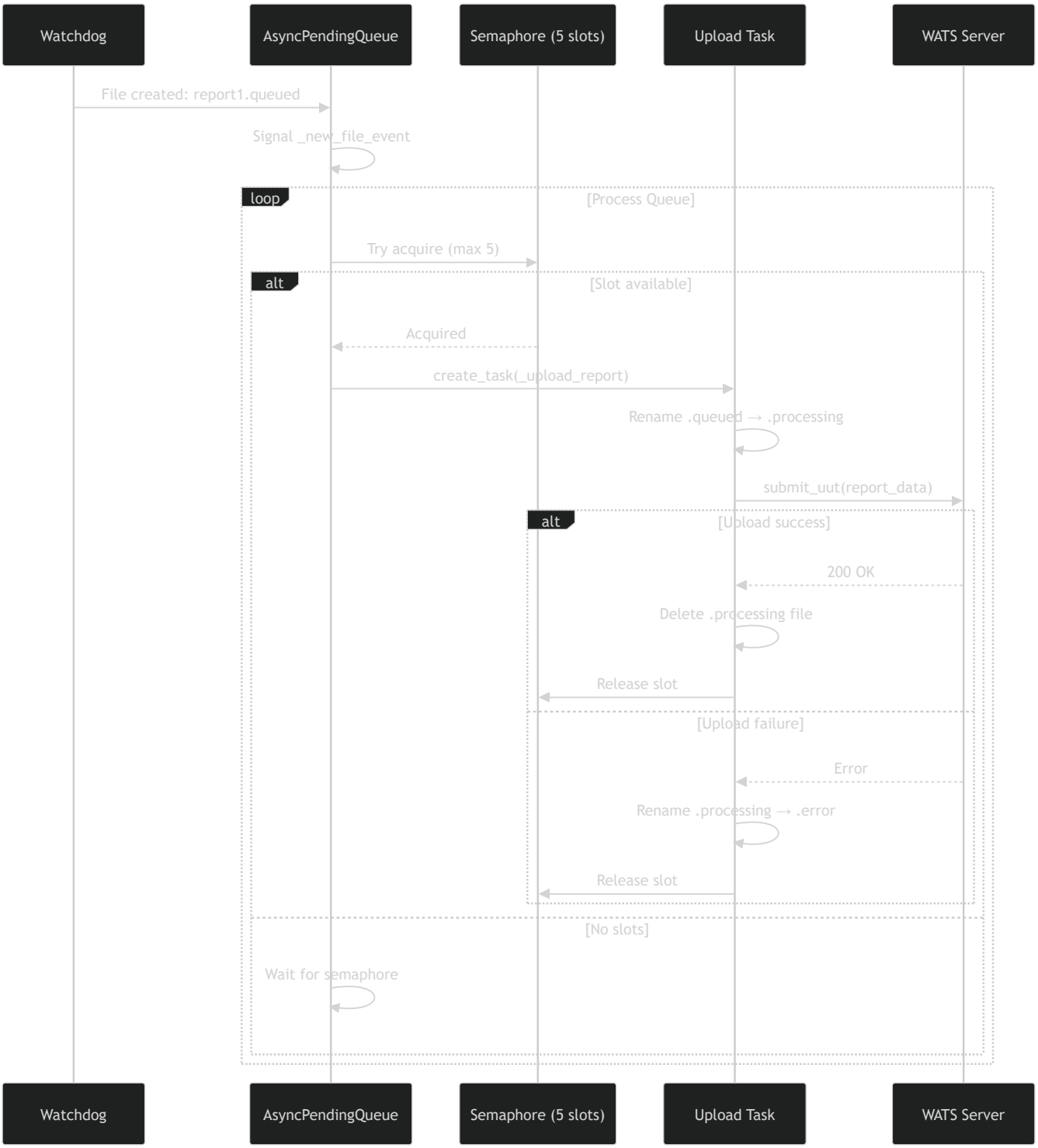

AsyncPendingQueue
Watchdog Observer
(File Monitor)
New file detected
Signal Event
asyncio.Semaphore
(Max 5 concurrent)
Rename
Acquire slot
Active Upload Tasks
Task 1
Task 2
Task 3
Task 4
Task 5
File System (reports_dir/)
*.processing
(Uploading)
Failure
Success
Pick next



File State Machine



Concurrent Upload Flow



Responsibilities

Responsibility	Implementation	Benefit
Concurrent Uploads	`asyncio.Semaphore(5)`	5 reports upload simultaneously

Responsibility	Implementation	Benefit
File Monitoring	`watchdog.Observer`	Instant detection of new reports
Retry Mechanism	`.error` files + delay	Auto-retry failed uploads
Crash Recovery	`.processing` timeout check	Reset stuck uploads
Queue Capacity	`max_queue_size` check	Prevent resource exhaustion
Graceful Shutdown	Wait for active tasks	No data loss on stop

Queue Comparison Matrix

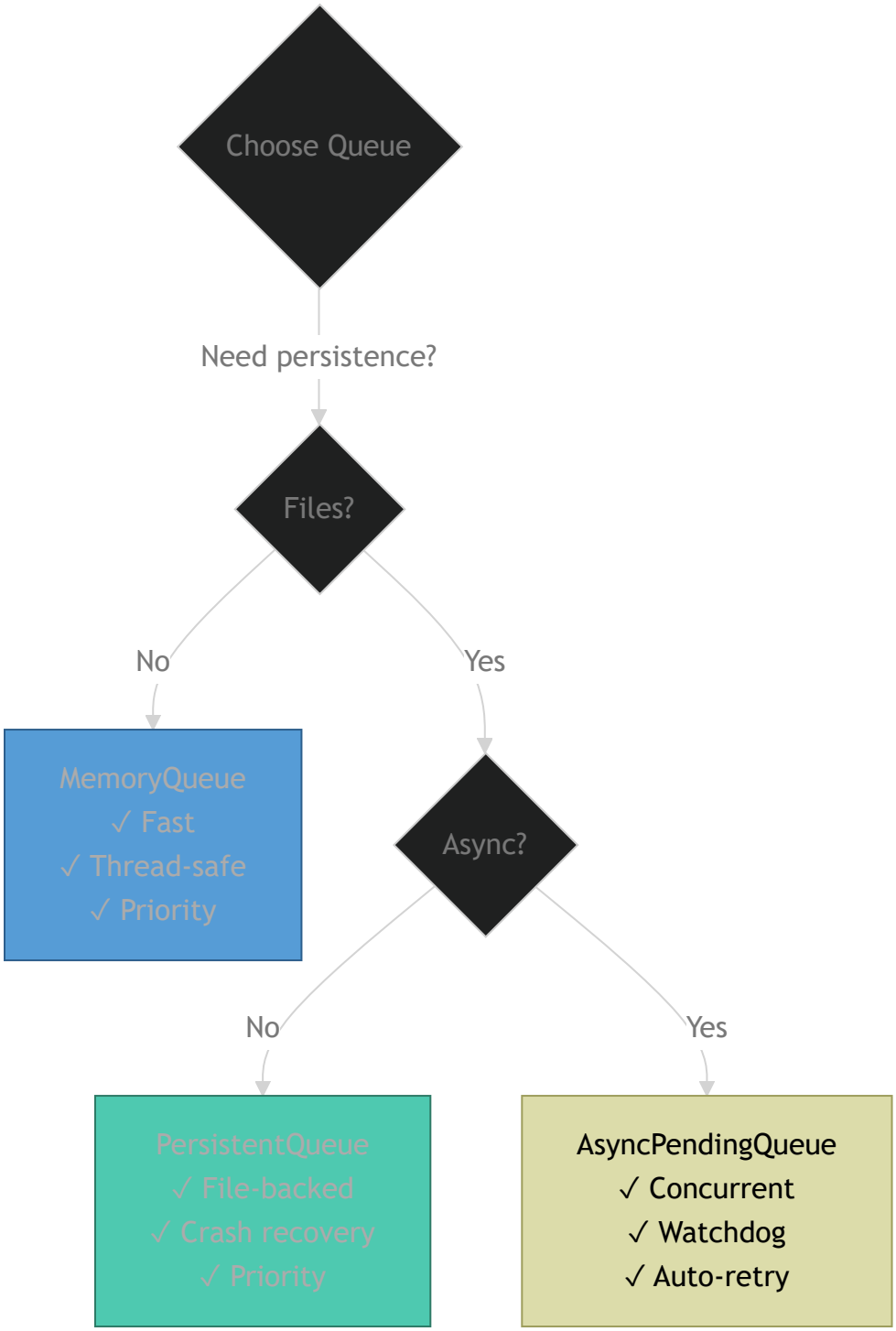
When to Use Which Queue

Feature	MemoryQueue	PersistentQueue	AsyncPendingQueue
Persistence	❌ No	✅ Yes (files)	✅ Yes (files)
Crash Recovery	❌ No	✅ Yes	✅ Yes
Thread Safety	✅ Yes	✅ Yes	⚠️ Single loop†
Async Support	⚠️ Via adapter	⚠️ Via adapter	✅ Native
Concurrent Processing	❌ No	❌ No	✅ Yes (N at a time)
Priority Queue	✅ Yes	✅ Yes	❌ No‡
File I/O	❌ None	✅ Atomic writes	✅ Async I/O
Watchdog Monitoring	❌ No	❌ No	✅ Yes
Retry Logic	✅ Via attempts	✅ Via attempts	✅ Auto-retry
Package	`pywats.queue`	`pywats_client.queue`	`pywats_client.service`

Notes:

- † AsyncPendingQueue is single-event-loop safe (asyncio), not thread-safe
- ‡ AsyncPendingQueue processes files alphabetically (FIFO), no priority

Use Case Recommendations



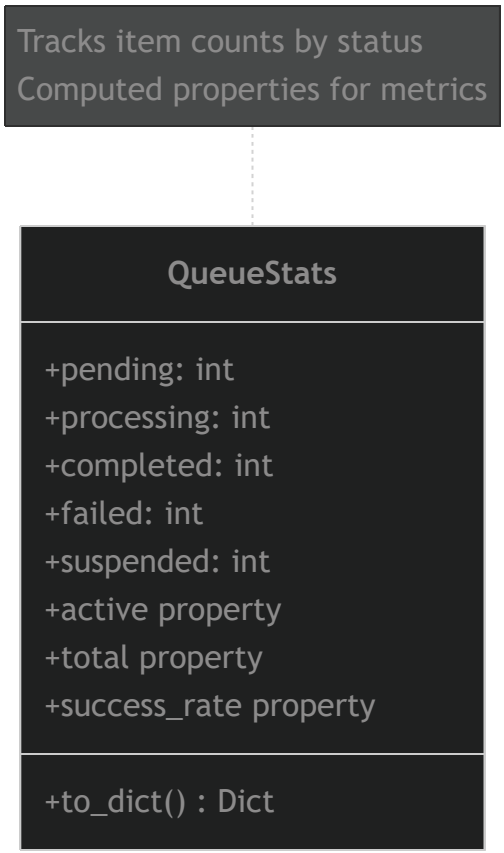
Decision Criteria:

1. **Choose MemoryQueue** when:
   - Quick in-process queuing
   - No persistence needed
   - Thread-safe priority queue required
   - Example: API request buffering

2. **Choose PersistentQueue** when:
    - File-based persistence required
    - Crash recovery important
    - Priority ordering needed
    - Sync (non-async) environment
    - Example: Offline report queue
3. **Choose AsyncPendingQueue** when:
    - Concurrent uploads required (5+ at a time)
    - Async environment (asyncio)
    - Watchdog file monitoring needed
    - Auto-retry important
    - Example: Background service uploads

Queue Statistics & Monitoring

QueueStats Model



Properties:

- `active = pending + processing + suspended`
- `total = pending + processing + completed + failed + suspended`
- `success_rate = completed / (completed + failed)` if any processed

Example Usage

```python
from pywats_client.queue import PersistentQueue

queue = PersistentQueue(queue_dir="C:/WATS/Queue")

# Get statistics
stats = queue.stats
print(f"Pending: {stats.pending}")
print(f"Processing: {stats.processing}")
print(f"Completed: {stats.completed}")
print(f"Failed: {stats.failed}")
print(f"Success Rate: {stats.success_rate:.1%}")

# Example output:
# Pending: 150
# Processing: 5
# Completed: 1200
# Failed: 8
# Success Rate: 99.3%
```

Queue Hooks (Extensibility)

Hook Interface

```python
from pywats.queue import QueueHooks, QueueItem

hooks = QueueHooks(
    on_add=lambda item: print(f"Added: {item.id}"),
    on_update=lambda item: print(f"Updated: {item.id} → {item.status.value}"),
    on_remove=lambda item_id: print(f"Removed: {item_id}")
)

queue = MemoryQueue(hooks=hooks)
```

Hook Events

Hook	Trigger	Use Case
on_add	`add()` called	Logging, metrics, notifications
on_update	`update()` called	Status change tracking, alerts
on_remove	`remove()` called	Cleanup, archival, metrics

Example: Prometheus Metrics

```python
from prometheus_client import Counter, Gauge

items_added = Counter('queue_items_added_total', 'Total items added')
items_completed = Counter('queue_items_completed_total', 'Total items completed')
queue_depth = Gauge('queue_depth', 'Current queue depth')

def on_add_hook(item):
    items_added.inc()
    queue_depth.inc()

def on_update_hook(item):
    if item.is_completed:
        items_completed.inc()
        queue_depth.dec()

hooks = QueueHooks(on_add=on_add_hook, on_update=on_update_hook)
queue = PersistentQueue(queue_dir="...", hooks=hooks)
```

Integration Examples

Example 1: Basic Report Queuing

```python
from pywats import pyWATS
from pywats_client.queue import PersistentQueue

# Initialize API and queue
api = pyWATS(service="https://python.wats.com", api_token="...")
queue = PersistentQueue(queue_dir="C:/WATS/Queue")

# Create and queue report
report = api.report.create_uut_report(
    serial_number="SN123456",
    part_number="PN789",
)
# ... add steps ...

# Add to queue with high priority
item = queue.add(report, priority=2)  # High priority
print(f"Queued: {item.id} with priority {item.priority}")

# Process queue
for item in queue.list_pending():
    item.mark_processing()
    queue.update(item)

    try:
        api.report.submit_uut(item.data)
        item.mark_completed()
    except Exception as e:
        if item.can_retry:
            item.mark_failed(str(e))
            # Will retry (attempts < max_attempts)
        else:
            item.mark_failed(str(e))
            # Max retries reached - permanent failure

    queue.update(item)
```

Example 2: Async Concurrent Uploads

```python
from pywats import AsyncWATS
from pywats_client.service import AsyncPendingQueue
import asyncio

async def main():
    # Initialize async API
    api = AsyncWATS(service="https://python.wats.com", api_token="...")

    # Start async upload queue (5 concurrent uploads)
    queue = AsyncPendingQueue(
        api=api,
        reports_dir="C:/WATS/Queue",
        max_concurrent=5,
        max_queue_size=10000
    )

    # Run queue in background
    queue_task = asyncio.create_task(queue.run())

    # Queue will automatically upload .queued files
    # Files can be dropped by converters or other processes

    # Monitor queue stats
    while True:
        stats = queue.stats
        print(f"Queued: {stats['queued_files']}, Active: {stats['active_uploads']}")
        await asyncio.sleep(10)

asyncio.run(main())
```

Example 3: Priority-Based Processing

```python
from pywats_client.queue import PersistentQueue

queue = PersistentQueue(queue_dir="C:/WATS/Queue")

# Add reports with different priorities
queue.add(report_critical, priority=1)   # Emergency - process first
queue.add(report_normal, priority=5)     # Normal - default
queue.add(report_bulk, priority=8)       # Low priority - process last

# Process in priority order
while queue.size > 0:
    item = queue.get_next()  # Gets highest priority item

    print(f"Processing {item.id} (priority: {item.priority})")
    item.mark_processing()
    queue.update(item)

    # ... process ...
```

Best Practices

1. Priority Assignment

Recommended Guidelines:

Priority	Category	Examples	Queue Time
1	Critical	Production line down, safety issues	< 1 minute
2-3	High	First article, debugging, urgent	< 5 minutes
4-6	Normal	Standard production tests	< 30 minutes
7-8	Low	Bulk uploads, re-submissions	< 2 hours
9-10	Background	Archives, cleanup, migration	When idle

2. Retry Strategy

Set max_attempts based on failure type:

```
# Transient network errors - retry more
queue.add(report, max_attempts=5)

# Validation errors - fail fast
queue.add(report, max_attempts=1)

# Default - balanced
queue.add(report, max_attempts=3)
```

3. Queue Capacity Management

Monitor queue depth:

```
MAX_QUEUE_DEPTH = 10000

if queue.size > MAX_QUEUE_DEPTH:
    logger.warning(f"Queue depth high: {queue.size}")
    # Option 1: Pause new submissions
    # Option 2: Increase upload concurrency
    # Option 3: Alert administrators
```

4. Error Handling

Distinguish error types:

```python
try:
    api.report.submit_uut(item.data)
    item.mark_completed()
except ValidationError as e:
    # Permanent error - don't retry
    item.max_attempts = 0
    item.mark_failed(f"Validation error: {e}")
except NetworkError as e:
    # Transient error - retry
    item.mark_failed(f"Network error: {e}")

queue.update(item)
```

5. Crash Recovery Testing

Test recovery scenarios:

```python
# 1. Simulate crash during processing
queue = PersistentQueue(queue_dir="...")
item = queue.get_next()
item.mark_processing()
queue.update(item)
# ... simulate crash (exit without completing) ...

# 2. Restart queue
queue2 = PersistentQueue(queue_dir="...")
# Item should be reset to PENDING with attempts incremented
```

Troubleshooting

Common Issues

1. Queue Not Processing

Symptoms: Items stay in PENDING state

Causes:

- No worker calling `get_next()`
- All semaphore slots full (AsyncPendingQueue)
- Queue paused/stopped

Fix:

```
# Check queue state
print(f"Size: {queue.size}")
print(f"Stats: {queue.stats}")

# For AsyncPendingQueue
print(f"State: {queue.state}")
print(f"Active uploads: {queue.stats['active_uploads']}")
```

2. Items Stuck in PROCESSING

Symptoms: Items never complete

Causes:

- Crash during processing
- Timeout not configured
- Forgot to call `update()`

Fix:

```
# PersistentQueue - restart (auto-recovery)
queue = PersistentQueue(queue_dir="...")  # Resets PROCESSING → PENDING

# Manual reset
item = queue.get("stuck-item-id")
item.reset_to_pending()
queue.update(item)
```

3. Priority Not Working

Symptoms: Low-priority items processed before high-priority

Cause: Using AsyncPendingQueue (no priority support)

Fix: Use PersistentQueue instead

```
# Wrong: AsyncPendingQueue ignores priority
queue = AsyncPendingQueue(...)  # FIFO only

# Right: PersistentQueue respects priority
queue = PersistentQueue(...)  # Priority + FIFO
```

4. High Memory Usage

Symptoms: Memory grows over time

Causes:

- Not deleting completed items
- Large `data` payloads
- Too many items in queue

Fix:

```
# Enable auto-delete completed items
queue = PersistentQueue(queue_dir="...", delete_completed=True)

# Manually clear completed
queue.clear_completed()

# Limit queue size
queue = PersistentQueue(queue_dir="...", max_size=10000)
```

Related Documentation

- Client & Service Architecture - Overall system design
- Converter Development Guide - Creating custom converters
- API Reference - Queue API documentation
- Installation Guide - Setup instructions

Key Takeaway: The queue system uses a **layered architecture** where base in-memory queues (`MemoryQueue`) are extended for persistence (`PersistentQueue`) and async concurrency

(`AsyncPendingQueue`). **Priority queuing** ensures critical reports are processed first, while **crash recovery** and **retry mechanisms** ensure reliability in production environments.